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COGNITIVE EMBODIED SOCIAL AGENTS RESEARCH

Modeling Bounded Rationality in Drug Shortage Pharmacists Using Attention-Guided Dynamic Decomposition

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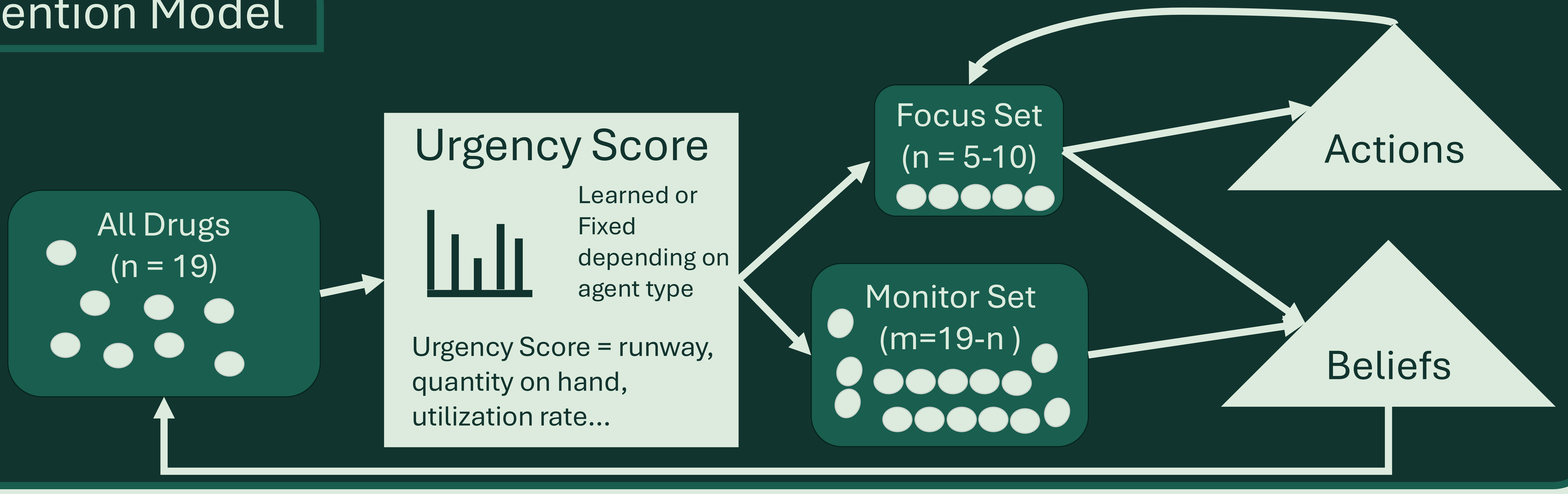
Abstract

Hospital pharmacists make high-stakes decisions to mitigate drug shortages under uncertainty, time pressure, and patient risk. Interviews revealed that pharmacists focus attention on a small subset of drugs, limiting cognitive effort to the most urgent cases. Motivated by these findings, we formalize a bounded rational, attention-guided decision framework that dynamically decomposes drugs into a subset for high-cost reasoning and a complementary subset for low-cost monitoring. We develop two agents: an Expert Agent that applies attention weights derived from pharmacist interviews, and a Learner Agent that adapts attention allocation over time through experience. Across simulated scenarios spanning short to long horizons, we show that attention-guided planning supports stable decision making without complete state reasoning. These results suggest that a primary decision is not what action to take, but where to allocate cognitive effort, and that attention-guided, satisficing strategies can reduce problem complexity while maintaining stable performance.

Attention Model



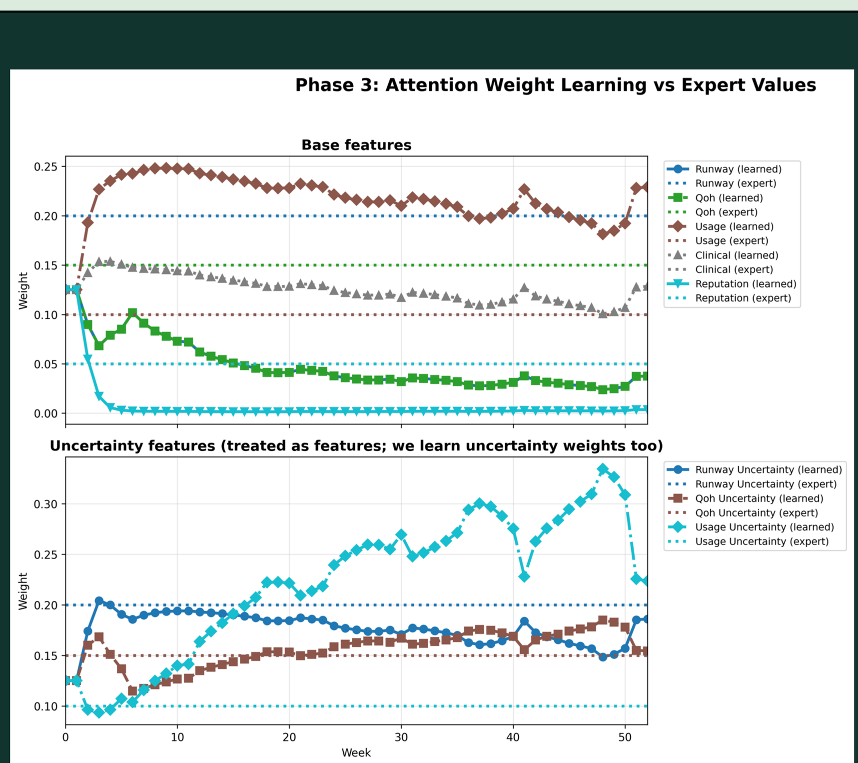
Try It Yourself!

QR placeholder

	A	B	C	D	E	F
Quantity On Hand	20ml/ 0.25	17 fl oz/0.9	10g/ 0.3	123 vials/ 0.6	55 packs/ 0.625	1000lbs / 0.1
Utilization Rate	15ml/ 0.5	3 fl oz/ 0.1	1g/ 0.2	10 vials / 0.8	5 packs/ 0.35	200lbs/ 0.95
Clinical Impact	1	5	3	6	2	4
Resupply weeks	1/ 0.95	3/ 0.5	7/ 0.25	3/ 0.1	11/ 0.3	6 / 0.8

Results

Learner Agent was able to learn its own betas for attention values



Takeaway

The first decision is not what action to taken; it is where to spend reasoning effort.